HCLTech | Future Ready Data Science Hackathon

IT Support Ticket Classification and Analytics Dashboard

Problem Statement and Participant Brief

Use case	IT Support Ticket Classification and Analytics Dashboard
Industry	Data Science / IT Service Management

1. Problem Statement

Organizations receive large volumes of IT support tickets related to hardware, software, networks, access and other technical issues. Manual categorization can delay assignment to the appropriate support team and make it difficult to identify recurring problems and changing support demand.

Build a machine-learning solution that automatically classifies tickets from their descriptions and provides an analytics dashboard to visualize ticket categories, recurring issues, support trends and areas requiring operational attention.

2. Functional Requirements

D	Capability	Requirement
FR-01	Ticket-data preparation	Prepare ticket descriptions and available metadata, handle missing or noisy text and document any label-quality issues.
FR-02	Multi-class classification	Predict an approved ticket category, such as hardware, software, network or access, together with a confidence score.
FR-03	Routing support	Map the predicted category to the appropriate support queue or assignment group where mapping data is supplied.
FR-04	Low-confidence handling	Flag uncertain, ambiguous or out-of-scope tickets for manual review rather than forcing a category.
FR-05	Recurring-issue analysis	Identify repeated themes or clusters within and across categories and show representative ticket descriptions.
FR-06	Analytics dashboard	Visualize category distribution, ticket volume over time, recurring issues, confidence, misclassification patterns and available resolution or handling trends.
FR-07	Model evaluation	Report per-class precision, recall and F1, macro and weighted averages, a confusion matrix and performance on minority categories.
FR-08	Explainability and reproducibility	Provide interpretable category signals or examples and document preprocessing, data split, model settings and assumptions.

3. Expected Submission

- **Classification prototype:** A working notebook, application or API that accepts ticket descriptions and returns predicted categories with confidence.
- **Analytics dashboard:** A dashboard showing category volumes, trends, recurring issues, model confidence and selected operational insights supported by the supplied data.
- **Evaluation report:** A concise report covering class distribution, split strategy, baseline comparison, per-class metrics, confusion matrix, error analysis and limitations.
- **Worked examples:** Examples covering common categories, an ambiguous ticket, a minority category and at least one low-confidence case.
- **Technical documentation:** A README with setup, input schema, category taxonomy, preprocessing, model approach and reproducibility guidance.

4. Evaluation Guidance

Dimension	What good looks like
Per-class classification quality	Report precision, recall and F1 for each category so strong majority-class performance does not hide weak minority-category results.
Macro and weighted performance	Use macro F1 to reflect performance across categories and weighted F1 to summarize volume-adjusted performance.
Confusion analysis	Explain the most frequent category confusions and show representative examples rather than reporting only aggregate scores.
Low-confidence behaviour	Evaluate whether uncertain tickets are correctly routed for review and whether confidence is reasonably aligned with correctness.
Recurring-issue usefulness	Verify that identified themes represent coherent recurring problems and link them to supporting ticket examples.
Dashboard accuracy	Confirm that filters, counts, trends and charts match the underlying prepared dataset and do not double-count records.

5. Scope Boundaries

- The classification taxonomy should be limited to labels available or explicitly defined for the hackathon dataset.
- The solution should assist categorization and routing; automatic closure, remediation and production ITSM changes are outside scope.
- Synthetic or de-identified tickets may be used, but generated labels, timestamp assumptions and assignment mappings must be identified.
- The candidate may choose the text representation, classification model, recurring-issue method and dashboard framework.
- Sensitive ticket content must be masked or excluded from dashboard views and model explanations where appropriate.